

Cancer and Blood Concentrations of the Co-mutagen Harmane in Essential Tremor

.

Background Blood concentrations of harmane, a tremor-producing neurotoxin, are elevated in essential tremor (ET). Harmane is also a co-mutagen. Objective To compare the prevalence of cancer in ET cases vs. controls, and determine whether blood harmane concentrations are elevated among ET cases with cancer. Methods Case-control design. Results 66/267 (24.7%) ET cases vs. 55/331 (16.6%) controls had cancer (adjusted OR 1.52, 95% CI 1.01 – 2.30, $p = 0.04$). Among specific cancer types, colon cancer was more prevalent in ET cases than controls (2.6% vs. 0.6%, $p = 0.04$). Log blood harmane concentration was higher in ET cases vs. controls ($p = 0.02$) and in participants with vs. without cancer ($p = 0.02$). Log blood harmane concentration was highest in ET cases with cancer when compared with other groups ($p = 0.009$). Discussion These links between cancer and ET and between high blood harmane and cancer in ET deserve further study.